



```
# Display data
df
```

|   | Name    | Age | Gender | Marks |
|---|---------|-----|--------|-------|
| 0 | Jai     | 17  | 0.0    | 90.0  |
| 1 | Princi  | 17  | 1.0    | 76.0  |
| 2 | Gaurav  | 18  | 0.0    | 75.2  |
| 3 | Anuj    | 17  | 0.0    | 74.0  |
| 4 | Ravi    | 18  | 0.0    | 65.0  |
| 5 | Natasha | 17  | 1.0    | 75.2  |
| 6 | Riya    | 17  | 1.0    | 71.0  |

## Filtering data in Data Wrangling

```
# Filter top scoring students
df = df[df['Marks'] >= 75].copy()

# Remove age column from filtered DataFrame
df.drop('Age', axis=1, inplace=True)

# Display data
df
```

|   | Name    | Gender | Marks |
|---|---------|--------|-------|
| 0 | Jai     | 0.0    | 90.0  |
| 1 | Princi  | 1.0    | 76.0  |
| 2 | Gaurav  | 0.0    | 75.2  |
| 5 | Natasha | 1.0    | 75.2  |

## Data Wrangling Using Merge Operation

```
# Import module
import pandas as pd

# Creating Dataframe
details = pd.DataFrame({
    'ID': [101, 102, 103, 104, 105,
          106, 107, 108, 109, 110],
    'NAME': ['Jagroop', 'Praveen', 'Harjot',
             'Pooja', 'Rahul', 'Nikita',
             'Saurabh', 'Ayush', 'Dolly', 'Mohit'],
    'BRANCH': ['CSE', 'CSE', 'CSE', 'CSE', 'CSE',
              'CSE', 'CSE', 'CSE', 'CSE', 'CSE']})

# Creating Dataframe
fees_status = pd.DataFrame(
    {'ID': [101, 102, 103, 104, 105,
            106, 107, 108, 109, 110],
     'PENDING': ['5000', '250', 'NIL',
                  '9000', '15000', 'NIL',
                  '4500', '1800', '250', 'NIL']})

# Merging Dataframe
print(pd.merge(details, fees_status, on='ID'))
```

|   | ID  | NAME    | BRANCH | PENDING |
|---|-----|---------|--------|---------|
| 0 | 101 | Jagroop | CSE    | 5000    |
| 1 | 102 | Praveen | CSE    | 250     |
| 2 | 103 | Harjot  | CSE    | NIL     |
| 3 | 104 | Pooja   | CSE    | 9000    |
| 4 | 105 | Rahul   | CSE    | 15000   |
| 5 | 106 | Nikita  | CSE    | NIL     |
| 6 | 107 | Saurabh | CSE    | 4500    |
| 7 | 108 | Ayush   | CSE    | 1800    |
| 8 | 109 | Dolly   | CSE    | 250     |
| 9 | 110 | Mohit   | CSE    | NIL     |

## Creating First Dataframe to Perform Merge Operation using Data Wrangling:

### Creating Second Dataframe to Perform Merge operation using Data Wrangling:

## Data Wrangling Using Merge Operation

|   | ID  | NAME    | BRANCH | PENDING |
|---|-----|---------|--------|---------|
| 0 | 101 | Jagroop | CSE    | 5000    |
| 1 | 102 | Praveen | CSE    | 250     |
| 2 | 103 | Harjot  | CSE    | NIL     |
| 3 | 104 | Pooja   | CSE    | 9000    |
| 4 | 105 | Rahul   | CSE    | 15000   |
| 5 | 106 | Nikita  | CSE    | NIL     |
| 6 | 107 | Saurabh | CSE    | 4500    |
| 7 | 108 | Ayush   | CSE    | 1800    |
| 8 | 109 | Dolly   | CSE    | 250     |
| 9 | 110 | Mohit   | CSE    | NIL     |

## Creating dataframe to use Grouping methods[Car selling datasets]

[illegible]

```

        'Toyota', 'Mahindra', 'Mahindra',
        'Ford', 'Toyota', 'Ford'],
    'Year': [2010, 2011, 2009, 2013,
             2010, 2011, 2011, 2010,
             2013, 2010, 2010, 2011],
    'Sold': [6, 7, 9, 8, 3, 5,
             2, 8, 7, 2, 4, 2]}

```

```

# Creating Dataframe of car_selling_data
df = pd.DataFrame(car_selling_data)

```

```

# printing Dataframe
print(df)

```

|    | Brand    | Year | Sold |
|----|----------|------|------|
| 0  | Maruti   | 2010 | 6    |
| 1  | Maruti   | 2011 | 7    |
| 2  | Maruti   | 2009 | 9    |
| 3  | Maruti   | 2013 | 8    |
| 4  | Hyundai  | 2010 | 3    |
| 5  | Hyundai  | 2011 | 5    |
| 6  | Toyota   | 2011 | 2    |
| 7  | Mahindra | 2010 | 8    |
| 8  | Mahindra | 2013 | 7    |
| 9  | Ford     | 2010 | 2    |
| 10 | Toyota   | 2010 | 4    |
| 11 | Ford     | 2011 | 2    |

#### Creating Dataframe to use Grouping methods[DATA OF THE YEAR 2010]

```

# Import module
import pandas as pd

# Creating Data
car_selling_data = {'Brand': ['Maruti', 'Maruti', 'Maruti',
                              'Maruti', 'Hyundai', 'Hyundai',
                              'Toyota', 'Mahindra', 'Mahindra',
                              'Ford', 'Toyota', 'Ford'],
                    'Year': [2010, 2011, 2009, 2013,
                             2010, 2011, 2011, 2010,
                             2013, 2010, 2010, 2011],
                    'Sold': [6, 7, 9, 8, 3, 5,
                             2, 8, 7, 2, 4, 2]}

```

```

# Creating Dataframe for Provided Data
df = pd.DataFrame(car_selling_data)

```

```

# Group the data when year = 2010
grouped = df.groupby('Year')
print(grouped.get_group(2010))

```

|    | Brand    | Year | Sold |
|----|----------|------|------|
| 0  | Maruti   | 2010 | 6    |
| 4  | Hyundai  | 2010 | 3    |
| 7  | Mahindra | 2010 | 8    |
| 9  | Ford     | 2010 | 2    |
| 10 | Toyota   | 2010 | 4    |

#### Creating a Student Dataset who want to participate in the event

```

# Import module
import pandas as pd

# Initializing Data
student_data = {'Name': ['Amit', 'Praveen', 'Jagroop',
                        'Rahul', 'Vishal', 'Suraj',
                        'Rishab', 'Satyapal', 'Amit',
                        'Rahul', 'Praveen', 'Amit'],
                'Roll_no': [23, 54, 29, 36, 59, 38,
                           12, 45, 34, 36, 54, 23],
                'Email': ['xxxx@gmail.com', 'xxxxxx@gmail.com',
                          'xxxxxx@gmail.com', 'xx@gmail.com',
                          'xxxx@gmail.com', 'xxxxx@gmail.com',
                          'xxxxx@gmail.com', 'xxxxx@gmail.com',
                          'xxxxx@gmail.com', 'xxxxxx@gmail.com',
                          'xxxxxxxx@gmail.com', 'xxxxxxxxx@gmail.com']}

# Creating Dataframe of Data
df = pd.DataFrame(student_data)

```

```
# Printing Dataframe
print(df)
```

|    | Name     | Roll_no | Email                                                          |
|----|----------|---------|----------------------------------------------------------------|
| 0  | Amit     | 23      | <a href="mailto:xxxx@gmail.com">xxxx@gmail.com</a>             |
| 1  | Praveen  | 54      | <a href="mailto:xxxxxx@gmail.com">xxxxxx@gmail.com</a>         |
| 2  | Jagroop  | 29      | <a href="mailto:xxxxxx@gmail.com">xxxxxx@gmail.com</a>         |
| 3  | Rahul    | 36      | <a href="mailto:xx@gmail.com">xx@gmail.com</a>                 |
| 4  | Vishal   | 59      | <a href="mailto:xxxx@gmail.com">xxxx@gmail.com</a>             |
| 5  | Suraj    | 38      | <a href="mailto:xxxxx@gmail.com">xxxxx@gmail.com</a>           |
| 6  | Rishab   | 12      | <a href="mailto:xxxxx@gmail.com">xxxxx@gmail.com</a>           |
| 7  | Satyapal | 45      | <a href="mailto:xxxxx@gmail.com">xxxxx@gmail.com</a>           |
| 8  | Amit     | 34      | <a href="mailto:xxxxx@gmail.com">xxxxx@gmail.com</a>           |
| 9  | Rahul    | 36      | <a href="mailto:xxxxxx@gmail.com">xxxxxx@gmail.com</a>         |
| 10 | Praveen  | 54      | <a href="mailto:xxxxxxxxxx@gmail.com">xxxxxxxxxx@gmail.com</a> |
| 11 | Amit     | 23      | <a href="mailto:xxxxxxxxxx@gmail.com">xxxxxxxxxx@gmail.com</a> |

Removing Duplicate data from the Dataset using Data wrangling

```
# import module
import pandas as pd

# initializing Data
student_data = {'Name': ['Amit', 'Praveen', 'Jagroop',
                        'Rahul', 'Vishal', 'Suraj',
                        'Rishab', 'Satyapal', 'Amit',
                        'Rahul', 'Praveen', 'Amit'],

                'Roll_no': [23, 54, 29, 36, 59, 38,
                           12, 45, 34, 36, 54, 23],

                'Email': ['xxxx@gmail.com', 'xxxxxx@gmail.com',
                          'xxxxxx@gmail.com', 'xx@gmail.com',
                          'xxxx@gmail.com', 'xxxxxx@gmail.com',
                          'xxxxxx@gmail.com', 'xxxxxx@gmail.com',
                          'xxxxxx@gmail.com', 'xxxxxx@gmail.com',
                          'xxxxxxxxxx@gmail.com', 'xxxxxxxxxx@gmail.com']}

# creating dataframe
df = pd.DataFrame(student_data)

# Here df.duplicated() list duplicate Entries in Rollno.
# So that ~(NOT) is placed in order to get non duplicate values.
non_duplicate = df[~df.duplicated('Roll_no')]

# printing non-duplicate values
print(non_duplicate)
```

|   | Name     | Roll_no | Email                                                  |
|---|----------|---------|--------------------------------------------------------|
| 0 | Amit     | 23      | <a href="mailto:xxxx@gmail.com">xxxx@gmail.com</a>     |
| 1 | Praveen  | 54      | <a href="mailto:xxxxxx@gmail.com">xxxxxx@gmail.com</a> |
| 2 | Jagroop  | 29      | <a href="mailto:xxxxxx@gmail.com">xxxxxx@gmail.com</a> |
| 3 | Rahul    | 36      | <a href="mailto:xx@gmail.com">xx@gmail.com</a>         |
| 4 | Vishal   | 59      | <a href="mailto:xxxx@gmail.com">xxxx@gmail.com</a>     |
| 5 | Suraj    | 38      | <a href="mailto:xxxxx@gmail.com">xxxxx@gmail.com</a>   |
| 6 | Rishab   | 12      | <a href="mailto:xxxxx@gmail.com">xxxxx@gmail.com</a>   |
| 7 | Satyapal | 45      | <a href="mailto:xxxxx@gmail.com">xxxxx@gmail.com</a>   |
| 8 | Amit     | 34      | <a href="mailto:xxxxx@gmail.com">xxxxx@gmail.com</a>   |

Creating New Datasets Using the Concatenation of Two Datasets In Data Wrangling.

```
# importing pandas module
import pandas as pd

# Define a dictionary containing employee data
data1 = {'Name': ['Jai', 'Princi', 'Gaurav', 'Anuj'],
         'Age': [27, 24, 22, 32],
         'Address': ['Nagpur', 'Kanpur', 'Allahabad', 'Kannuaj'],
         'Qualification': ['Msc', 'MA', 'MCA', 'Phd'],
         'Mobile No': [97, 91, 58, 76]}

# Define a dictionary containing employee data
data2 = {'Name': ['Gaurav', 'Anuj', 'Dhiraj', 'Hitesh'],
         'Age': [22, 32, 12, 52],
         'Address': ['Allahabad', 'Kannuaj', 'Allahabad', 'Kannuaj'],
         'Qualification': ['MCA', 'Phd', 'Bcom', 'B.hons'],
         'Salary': [1000, 2000, 3000, 4000]}

# Convert the dictionary into DataFrame
df = pd.DataFrame(data1, index=[0, 1, 2, 3])
```

```
# Convert the dictionary into DataFrame
df1 = pd.DataFrame(data2, index=[2, 3, 6, 7])
```

```
res = pd.concat([df, df1])
res
```

|   | Name   | Age | Address   | Qualification | Mobile No | Salary |
|---|--------|-----|-----------|---------------|-----------|--------|
| 0 | Jai    | 27  | Nagpur    | Msc           | 97.0      | NaN    |
| 1 | Princi | 24  | Kanpur    | MA            | 91.0      | NaN    |
| 2 | Gaurav | 22  | Allahabad | MCA           | 58.0      | NaN    |
| 3 | Anuj   | 32  | Kannuaj   | Phd           | 76.0      | NaN    |
| 2 | Gaurav | 22  | Allahabad | MCA           | NaN       | 1000.0 |
| 3 | Anuj   | 32  | Kannuaj   | Phd           | NaN       | 2000.0 |
| 6 | Dhiraj | 12  | Allahabad | Bcom          | NaN       | 3000.0 |
| 7 | Hitesh | 52  | Kannuaj   | B.hons        | NaN       | 4000.0 |

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